

Monthly Intelligence Report

Edition 003 — The DE-LU Border Quadrant Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

91

11 HV $\geq$ 110 kV · 15 MV (20–65 kV) ·
65 distribution-tier

GRID LINES MAPPED

676.9 km

198 HV $\geq$ 110 kV · 478 MV · OSM live
extraction

MC SIMULATIONS

1.82 M

20,000 per substation · Gaussian
copula

PUBLIC DATA SOURCES

22

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 6,000+ source data points per run, executes 1.82 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 0.04 billion arithmetic operations — producing 5,460 output data points with full uncertainty quantification across 91 substations (LIVE — OSM) and 628 grid lines spanning 676.9 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 22 verified public sources, making the methodology fully auditable and reproducible. Initially covering Luxembourg’s entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation Grevenmacher 001 (MV)

Grevenmacher, Grevenmacher · distribution-tier

0.349

Low

0.178

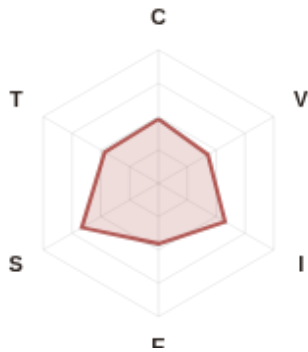
76th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.482 / 0.30 (161%)

I Infrastructure: 0.577 / 0.25 (231%)

S Saturation: 0.665 / 0.20 (332%)

V Voltage: 0.428 / 0.10 (428%)

E Economic: 0.454 / 0.10 (454%)

T Transition: 0.466 / 0.05 (933%)

MODIFIERS

R3 Consequence: 1.082 ↑ amplifies

R4 Graph Criticality: 1.238 ↑ amplifies

R6a Restoration: 0.939 ↓ dampens

R6b Seismic: 0.999 → neutral

R7 Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for June 2026 — is **Substation Grevenmacher 001 (MV)** in Grevenmacher, Grevenmacher. With an R_final of 0.349 (Low band, 76th fleet percentile), its risk profile is dominated by the T (Transition) component at 933% of its weight ceiling, followed by E (Economic) at 454%. Modifiers collectively shift R_base by 16.9%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Luxembourg Numérique / Luxembourg 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0000

Range [0.99, 1.05]

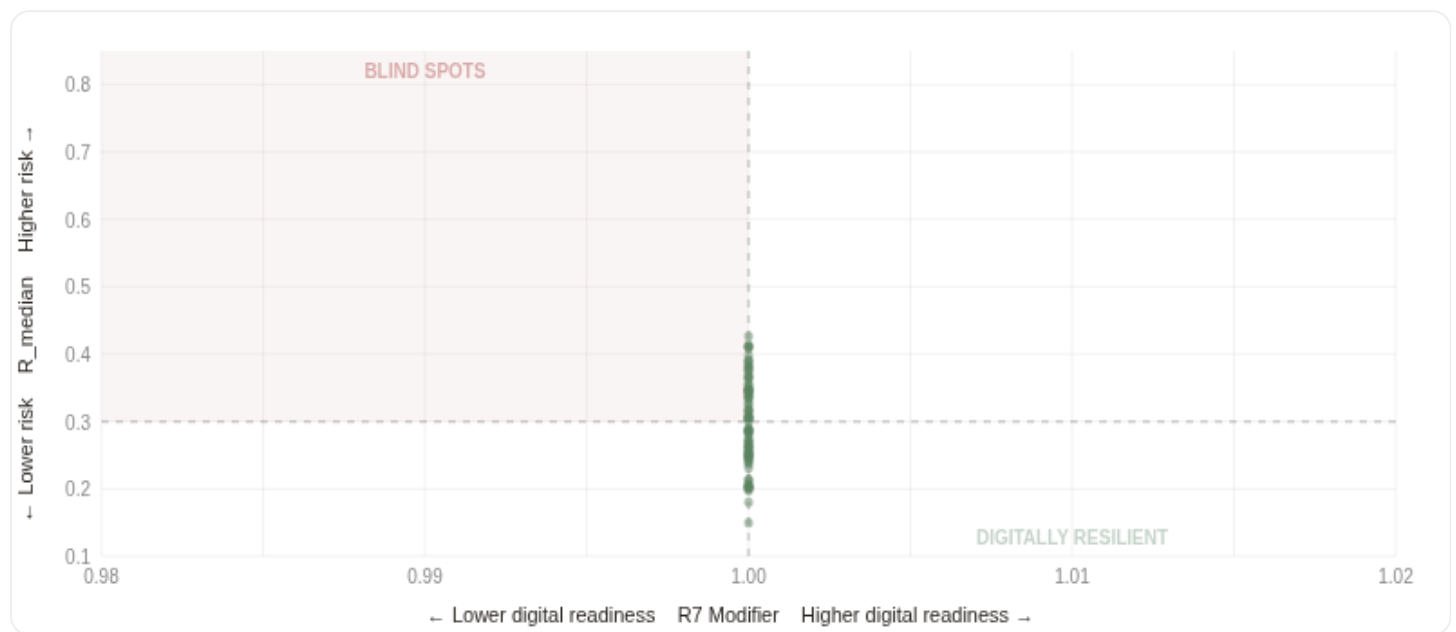
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.13

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in —. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier – derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Capital-Intensive / High-Consequence

Est. VoLL: €15-30/kWh

0 substations (0.0%) High/Critical: 0 (0%) Avg R: 0.000 Avg E: 0.000 / 0.10

Low 0

Med 0

High 0

Crit 0


Industrial / Medium–Large Enterprise

Est. VoLL: €8–15/kWh

34 substations (37.4%)
High/Critical: **0 (0.0%)**
Avg R: **0.334**
Avg E: **0.604 / 0.10**

Low 34

Med 0

High 0

Crit 0

24 Commercial / Mixed

Est. VoLL: €3–8/kWh

31 substations (34.1%) High/Critical: 0 (0.0%) Avg R: 0.243 Avg E: 0.481 / 0.10

Low 31 Med 0 High 0 Crit 0

Light-Rural / Agricultural

Est. VoLL: €1–3/kWh

26 substations (28.6%) High/Critical: 0 (0.0%) Avg R: 0.322 Avg E: 0.460 / 0.10

Low 26 Med 0 High 0 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric ($R3 \geq 1.14$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in — — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 6.3%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Canton Digital Readiness Ranking

Canton ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify Canton combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Capellen	1.0000
2	Clervaux	1.0000
3	Diekirch	1.0000
4	Echternach Δ high R	1.0000
5	Esch-sur-Alzette Δ high R	1.0000
6	Grevenmacher Δ high R	1.0000
7	Luxembourg Δ high R	1.0000
8	Mersch	1.0000
9	Redange	1.0000
10	Remich Δ high R	1.0000
11	Wiltz	1.0000

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Capellen** has the lowest average R7 (1.0000), while **Wiltz** leads at 1.0000 — a spread of 0.0000 across the R7 range. **0 region regions** combine below-median

digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

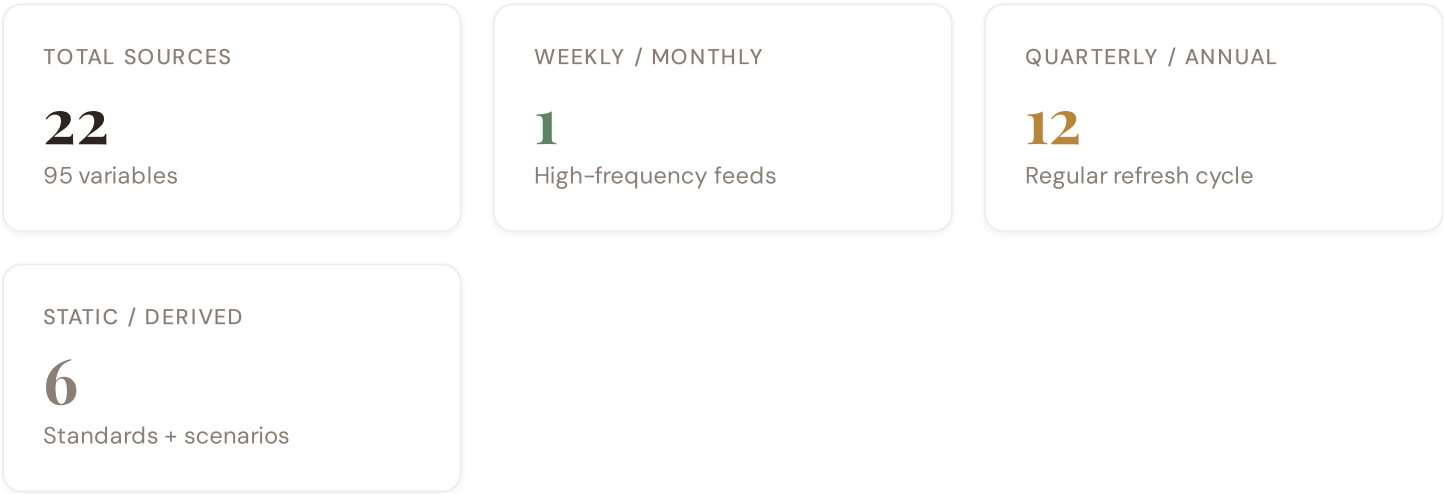
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0000, median = 1.0000; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

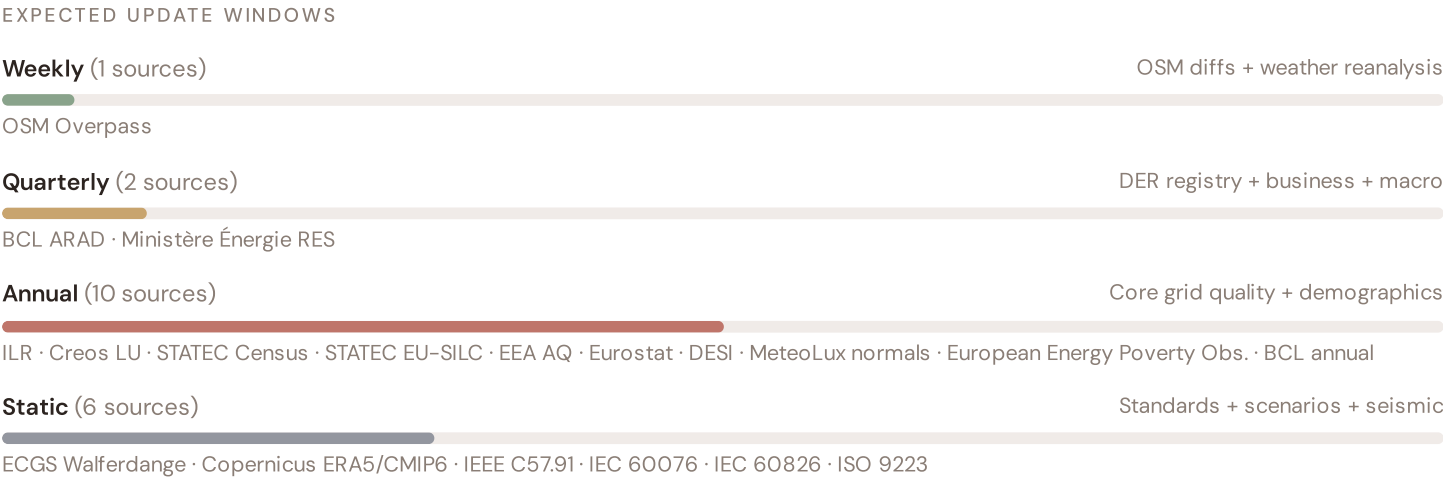
The SSI v4.0.2 draws on **22 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — June 2026

OSM	OpenStreetMap Power Infrastructure	CONTINUOUS	Node/edge	4 vars
BCL	Banque centrale du Luxembourg ARAD	QUARTERLY	Canton	2 vars
LUX-EN	Ministère de l'Énergie — RES register	QUARTERLY	National	3 vars
CREOS	Creos Luxembourg Annual Activity Report	ANNUAL	National	8 vars
ILR	ILR Quality of Supply Reports	ANNUAL	National	5 vars
STATEC	STATEC — Institut national de la statistique	ANNUAL	Canton	8 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km	2 vars
EUR	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy & Society Index	ANNUAL	EU Regional	1 var
EEPR	European Energy Poverty Observatory	ANNUAL	NUTS2	1 var
ECGS	ECGS Walferdange — European Center for Geodynamics & Seismology	STATIC	~5 km	2 vars
HAZARD	Administration de la gestion de l'eau — Flood maps	STATIC	~250 m	1 var
CTL	Cattenom NPP proximity decay (EDF)	STATIC	Canton	1 var
IEEE-1	IEEE C57.91 Thermal Model	STATIC	Asset-level	1 var
IEC-1	IEC 60076 Power Transformers	STATIC	Asset-level	1 var
IEC-826	IEC 60826 Overhead Line Loading	STATIC	Zone	1 var
ISO-9223	ISO 9223 Atmospheric Corrosion	DERIVED	Canton	1 var
EPEX	EPEX SPOT — DE-LU Bidding Zone	HOURLY	Zone	3 vars
ENTSOE	ENTSO-E Transparency Platform	HOURLY	Bidding zone	4 vars
METE0	MeteoLux — Findel Station	HOURLY/ANNUAL	Canton (interp.)	5 vars
CDS	Copernicus CDS / ERA5 + CMIP6 SSP2-4.5	HOURLY/STATIC	0.25° (~25 km)	4 vars
OPM	Open-Meteo Weather API	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **91 substation scores remain unchanged** this edition. The fleet median R of 0.300 (mean 0.299) reflects a distribution skewed toward the lower bands, with 100.0% in Low and 0.0% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

20 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio = STATEC EU-SILC)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection (NetworkX)	\$5

F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2–4.5) — pending live ERA5 file	§5
F6	ENHANCED	R7 Digital Readiness — Canton-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion (ISO 9223), hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — high-tech employment + financial-centre density (BCL + STATEC)	§6
LU1	NEW	Luxembourg onboarded — 10 source modules (no d02b: Creos LU single TSO+DSO), d05_osm LIVE (91 substations) P2	Stage 1
LU2	NEW	LU regulatory stack mapped: Creos LU / ILR / STATEC / BCL / MeteoLux / ECGS Walferdange P2	Stage 1
LU3	NEW	Scoring engine: 12 cantons × 6 components → R_final + Markov ETTC + 20k MC + Sobol P2	Stage 2
LU4	NEW	Dashboard: 8 HTML pages + Pattern C ssi-metadata.js + 22-source registry P2	Stage 3
LU5	ENHANCED	R3 economic tier thresholds LU-calibrated [1.110 / 1.075 / 1.020] from actual distribution P2	Stage 4
LU6	ENHANCED	R6b seismic α band LU-narrow [0.05, 0.20] (Ardennes-Eifel basement, narrower than CZ) P2	Stage 4
LU7	DESIGN	Cantons = L1 = L2 (12 cantons, no commune downscale — communes shown only for context) P2	Stage 4
LU8	ENHANCED	Markov rates younger-fleet calibration ($\lambda_{12}=0.035, \lambda_{23}=0.05, \lambda_{34}=0.08$ vs CZ baseline) P2	Stage 4
LU9	NEW	Automation pipeline scaffolded — FIRST_REFRESH=2026-07 — monthly + quarterly + annual workflows P2	Stage 5
LU10	NEW	Landing-page Luxembourg map-active (oecd → active flip) + OECD count update P2	Stage 6
D1	DATA	22-source LU registry verified (vs Italy 30, France 35, CZ 30) — all open-licensed	Data
D2	DATA	OSM Overpass live extraction: 91 substations + 628 power lines + 676.9 km	Data

SECTION D — MONTHLY DEEP-DIVE

The DE-LU Border Quadrant Corridor: Where Coal Reliance Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** DE-LU Border Quadrant corridor corridor · **002** DER saturation hotspots (border cantons) · **003** Climate vulnerability under CMIP6 · **004** Urban-Rural economic impact disparity · **005** coal-fleet decommissioning risk and transition risk · **006** Infrastructure age vs smart-meter rollout · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Canton Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the DE-LU Border Quadrant corridor, Why Now

DE-LU BORDER QUADRANT
CORRIDOR SUBSTATIONS

7
0 EHV · 7 HV

HIGH BAND

0
0.0% of corridor

CORRIDOR MEDIAN R

0.367
Fleet median: 0.300

WORST CANTON

Grevenmacher
Avg R: 0.374 · 7 substations

The Grevenmacher hosts **7 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High, with only **7** in the Low band. 100% of corridor substations score above the national fleet median of 0.300. The corridor median R of 0.367 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Poste Dupont	Grevenmacher	0.427	Low	0.473	0.451	0.562	0.462	0.635	0.535	
Poste Grevenmacher	Grevenmacher	0.411	Low	0.467	0.406	0.564	0.449	0.617	0.459	
Poste Weiergewan	Grevenmacher	0.391	Low	0.517	0.438	0.569	0.432	0.594	0.504	
Poste Findel	Grevenmacher	0.367	Low	0.460	0.417	0.599	0.444	0.586	0.467	
Substation Grevenmacher 001 (MV)	Grevenmacher	0.349	Low	0.482	0.428	0.577	0.454	0.665	0.466	
Poste Betzdorf	Grevenmacher	0.340	Low	0.503	0.413	0.609	0.479	0.588	0.545	
Poste Potaschbierg	Grevenmacher	0.335	Low	0.489	0.440	0.553	0.442	0.630	0.499	

... -3 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — GREVENMACHER VS FLEET

C — Continuity	0.484 vs 0.499 (-3%)
I — Infrastructure	0.576 vs 0.445 (+30%)
S — Saturation	0.616 vs 0.543 (+13%)
V — Voltage	0.427 vs 0.444 (-4%)
E — Economic	0.451 vs 0.521 (-13%)
T — Transition	0.496 vs 0.502 (-1%)

MODIFIER AVERAGES — GREVENMACHER VS FLEET

R3 Consequence	1.082	fleet 1.093	↑
R4 Graph Crit.	1.238	fleet 1.238	↑
R6a Restoration	0.939	fleet 0.957	↓
R6b Seismic	0.999	fleet 0.999	→
R7 Digital Read.	1.000	fleet 1.000	→

The dominant risk driver across the Grevenmacher corridor is **T (Transition)**, averaging 0.496 against a fleet average of 0.502 — occupying 993% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.451 vs fleet 0.521 (451% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.082 (fleet: 1.093), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.999, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Canton Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The region regions–level breakdown reveals significant intra–regional variation. **Grevenmacher** leads with a mean R of 0.374 across 7 substations, while **Grevenmacher** scores 0.374 — a spread of 0.000 within a single corridor. The SSI's substation–level granularity reveals that the Grevenmacher is not uniformly stressed but contains distinct sub–corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

O Grevenmacher substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Grevenmacher sub–corridor as the highest–priority target — where DER deployment is accelerating against ageing infrastructure with above–average continuity deficits and significant socio–economic exposure. The SSI's silo–breaking approach reveals what no single–discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

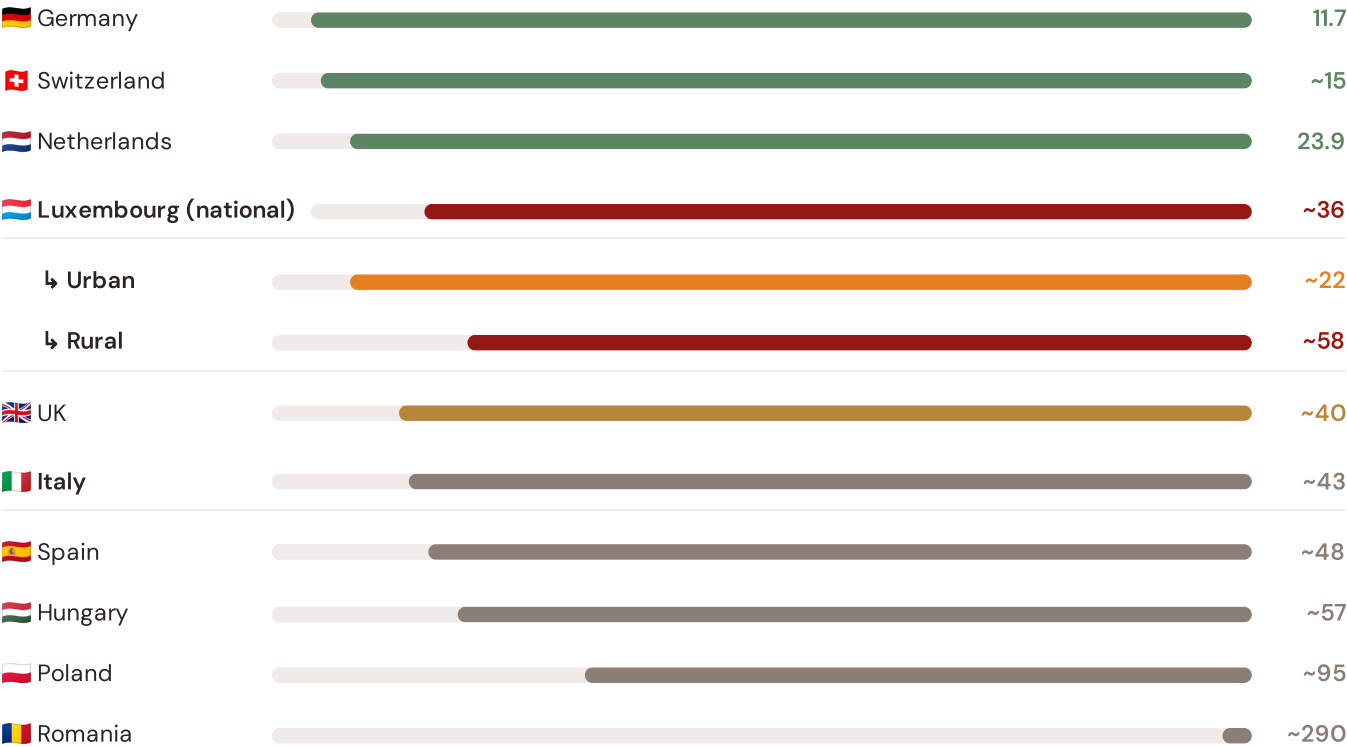
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep–dive theme. This month: **SAIDI comparison** — because the DE–LU Border Quadrant corridor analysis hinges on Luxembourg's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



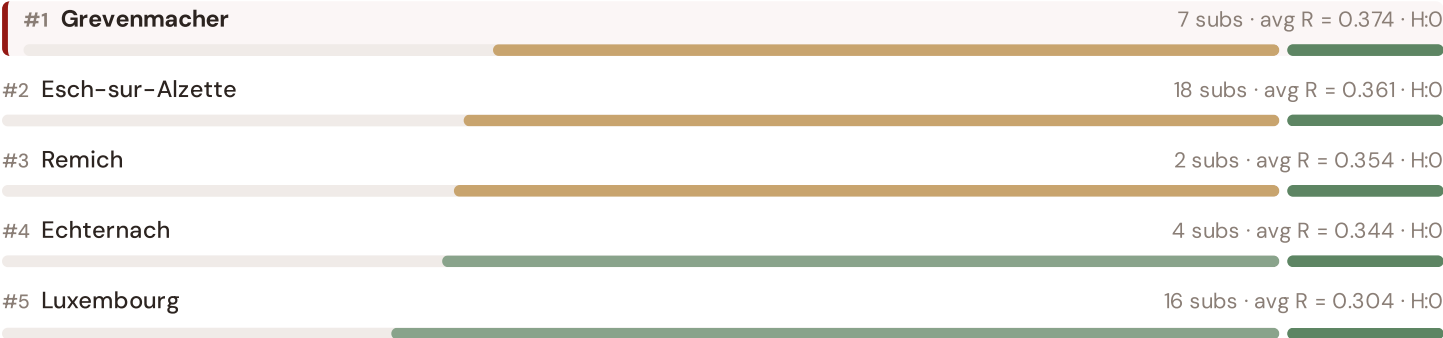
Source: CEER 7th Benchmarking Report (2022); ILR TIQE (2023); Bundesnetzagentur (2024). Luxembourg sub-categories from ILR territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

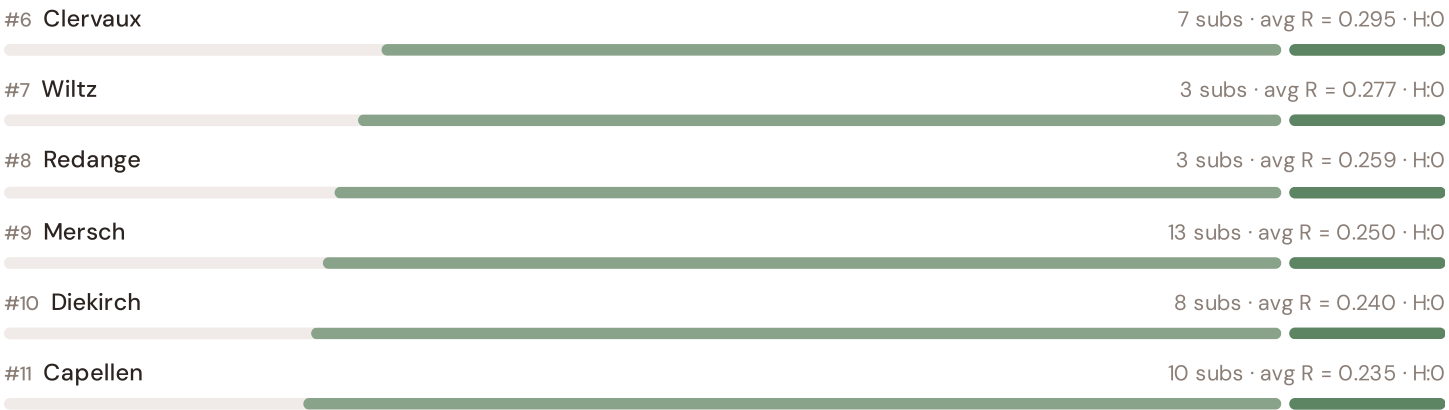
This month's key insight: The Grevenmacher corridor deep-dive shows **7 substations** (of 7) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.484 is **1.0× the fleet average** of 0.499. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 LU Cantons

Where does the DE-LU Border Quadrant corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



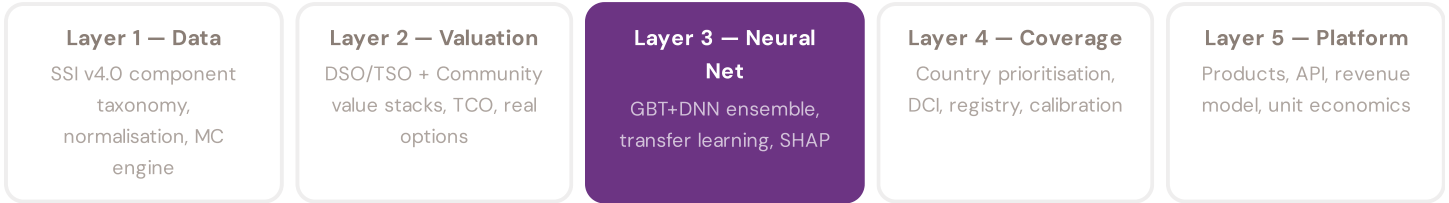


The Grevenmacher ranks **#1 of 11 region regions** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed are Grevenmacher, Esch-sur-Alzette, Remich, while the three most resilient are Capellen, Diekirch, Mersch. 5 of 11 score above the fleet average of 0.299. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Creos LU/Creos LU and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 91 substations with complete feature vectors. Average CI_width of 0.169 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 91 Low, 0 Medium, 0 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

BCL ARAD, Ministère Énergie RES are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Creos Luxembourg Annual Activity Report** (8 variables → C1 D1L, C2 N1L, E1 peak load, E2 growth, E3 DER ratio, T1 RES share (combined TSO+DSO single op)).

FLEET SPOTLIGHT

Capellen: Highest Risk Concentration

Capellen has the highest concentration of High/Critical substations: **0 of 10 (0%)** are in the upper risk bands — avg R = 0.235. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Luxembourg utility, ILR, Creos LU, Creos LU, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 91 substations (LIVE OSM) · ~628 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5

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